PRACTICE SHEET 1

1. The speed of a bus is 72 km/hr. The distance covered by the bus in 5 seconds is
(1) 100 m (2) 60 m
(3) 50 m (4) 74.5 m

2. Two men start together to walk a certain distance, one at 4 km/h and another at 3 km/h. The former arrives half an hour before the latter. Find the distance.
(1) 8 km (2) 7 km
(3) 6 km (4) 9 km

3. A train starts from a place A at 6 a.m. and arrives at another place B at 4.30 p.m. on the same day. If the speed of the train is 40 km per hour, find the distance travelled by the train?
(1) 420 km (2) 230 km
(3) 320 km (4) 400 km

4. A boy runs 20 km in 2.5 hours. How long will he take to run 32 km at double the previous speed?
(1) 2 hours (2) 2.5 hours
(3) 4.5hours (4) 5 hours

5. A train is moving with the speed of 180 km/hr. Its speed (in metres per second) is:
(1) 5 (2) 40
(3) 30 (4) 50

6. A man riding his bicycle covers 150 metres in 25 seconds. What is his speed in km per hour?
(1) 25 (2) 21.6
(3) 23 (4) 20

7. Walking at the rate of 4 km an hour, a man covers a certain distance in 3 hours 45 minutes. If he covers the same distance on cycle, cycling at the rate of 16·5 km/hour, the time taken by him is
(1) 55.45 minutes (2) 54.55 minutes
(3) 55.44 minutes (4) 45.55 minutes

8. A train covers a distance of 10 km in 12 minutes. If its speed is decreased by 5 km/hr, the time taken by it to cover the same distance will be:
(1) 10 minutes
(2) 13 minutes 20 sec
(3) 13 minutes
(4) 11 minutes 20 sec

9. A man rides at the rate of 18 km/ hr, but stops for 6 mins. to change horses at the end of every 7th km. The time that he will take to cover a distance of 90 km is:
(1) 6 hrs. (2) 6 hrs. 12 min.
(3) 6 hrs.18 min. (4) 6 hrs. 24 min.

10. A speed of 30.6 km/.hr is the same as
(1) 8.5 m/sec. (2) l0 m/sec.
(3) 12 m/sec. (4) 15.5 m/sec.

11. A man covers 2/15 of the total journey by train, 9/20 by bus and the remaining 10 km on foot. His total journey (in km) is
(1) 15.6 (2) 24
(3) 16.4 (4) 12.8

12. A man travels for 5 hours 15 minutes. If he covers the first half of the journey at 60 km/h and rest at 45 km/h. Find the total distance travelled by him.
(1) 1028.85 km. (2) 189 km.
(3) 378 km. (4) 270 km.

13. A car can finish a certain journey in 10 hours at the speed of 42 kmph. In order to cover the same distance in 7 hours, the speed of the car (km/h) must be increased by:
(1) 12 (2) 15
(3) 18 (4) 24

14. A man cycles at the speed of 8km/hr and reaches office at 11 am and when he cycles at the speed of 12 km/hr he reaches office at 9 am. At what speed should he cycle so that he reaches his office at 10 am?
(1) 9.6 kmph
(2) 10 kmph
(3) 11.2 kmph
(4) Cannot be determined

15. A bus travels at the speed of 36 km/hr, then the distance covered by it in one second is
(1) 10 metre (2) 15 metre
(3) 12.5 metre (4) 13.5 metre

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16. A boy rides his bicycle 10km at an average speed of 12 km/hr and again travels 12 km at an average speed of 10 km/hr. His average speed for the entire trip is approximately :
(1) 10.4 km/hr (2) 10.8 km/hr
(3) 11.0 km/hr (4) 12.2 km/hr

17. A person travels 600 km by train at 80km/hr, 800 km by ship at 40 km/hr 500 km by aeroplane at 400 km/hr and 100 km by car at 50km/hr. What is the average speed for the entire distance ?
(1) $65\frac{5}{123}$ vkm./hr. (2) 60 km./hr.
(3) $60\frac{5}{123}$ km./hr. (4) 62 km./hr.

18. A train moves with a speed of 30 kmph for 12 minutes and for next 8 minutes at a speed of 45 kmph. Find the average speed of the train:
(1) 37.5 kmph (2) 36 kmph
(3) 48 kmph (4) 30 kmph

19. A man covers half of his journey at 6km/hr and the remaining half at 3km/hr. His average speed is
(1) 9 km/hr (2) 4.5 km/hr
(3) 4 km/hr (4) 3 km/hr

20. A man goes from A to B at a uniform speed of 12 kmph and returns with a uniform speed of 4 kmph His average speed (in kmph) for the whole journey is :
(1) 8 (2) 7.5
(3) 6 (4) 4.5

21. A train covers a distance of 3584 km in 2 days 8 hours. If it covers 1440 km on the first day and 1608 km on the second day, by how much does the average speed of the train for the remaining part of the journey differ from that for the entire journey ?
(1) 3 km/hour more (2) 3 km/hour less
(3) 4 km/hour more (4) 5 km/hour less

22. A man travels a distance of 24 km at 6 kmph. Another distance of 24 km at 8 kmph and a third distance of 24 km at 12 kmph. His average speed for the whole journey (in kmph) is
(1) $8\frac{2}{3}$ (2) 8
(3) $2\frac{10}{13}$ (4) 9

23. A constant distance from Chennai to Bangalore is covered by Express train at 100 km/hr. If it returns to the same distance at 80 km/hr, then the average speed during the whole journey is
(1) 90.20 km/hr (2) 88.78 km/hr
(3) 88.98 km/hr (4) 88.89 km/hr

24. A person went from A to B at an average speed of x km/hr and returned from B to A at an average speed of y km/hr. What was his average speed during the total journey
(1) $\frac{x+y}{xy}$ (2) $\frac{2xy}{x+y}$
(3) $\frac{2}{x+y}$ (4) $\frac{1}{x}+\frac{1}{y}$

25. A man goes from Mysore to Bangalore at a uniform speed of 40 km/hr and comes back to Mysore at a uniform speed of 60 km/hr. His average speed for the whole journey is
(1) 48 km/hr (2) 50 km/hr
(3) 54 km/hr (4) 55 km/hr

26. A man goes from a place A to B at a speed of 12 km/hr and returns from B to A at a speed of 18 km/hr. The average speed for the whole journey is
(1) $14\frac{2}{5}$ km/hr (2) 15 km/hr
(3) $15\frac{1}{2}$ km/hr (4) 16 km/hr

27. One third of a certain journey is covered at the rate of 25 km/ hour, one-fourth at the rate of 30 km/hour and the rest at 50 km/ hour. The average speed for the whole journey is
(1) 35 km/hour (2) $33\frac{1}{3}$ km/hour
(3) 30 km/hour (4) $37\frac{1}{12}$ km/hour

28. A man completes 30 km of a journey at the speed of 6 km/hr and the remaining 40 km of the journey in 5 hours. His average speed for the whole journey is

(1) 7 km/hr (2) $6\frac{4}{11}$ km/hr
(3) 8 km/hr (4) 7.5 km/hr

29. A man covers the journey from a station A to station B at a uniform speed of 36 km/hr and returns to A with a uniform speed of 45 km/hr. His average speed for the whole journey is
(1) 40 km/hr (2) 40.5 km/hr
(3) 41 km/hr (4) 42 km/hr

30. The speed of a train going from Nagpur to Allahabad is 100 kmph while its speed is 150 kmph when coming back from Allahabad to Nagpur. Then the average speed during the whole journey is :
(1) 120 kmph (2) 125 kmph
(3) 140 kmph (4) 135 kmph

ANSWER KEY																			
1.	(1)	2.	(3)	3.	(1)	4.	(1)	5.	(4)	6.	(2)	7.	(2)	8.	(2)	9.	(2)	10.	(1)
11.	(2)	12.	(4)	13.	(3)	14.	(1)	15.	(1)	16.	(2)	17.	(1)	18.	(2)	19.	(3)	20.	(3)
21.	(1)	22.	(2)	23.	(4)	24.	(2)	25.	(1)	26.	(1)	27.	(2)	28.	(3)	29.	(1)	30.	(1)

PRACTICE SHEET 2

1. The length of a train and that of a platform are equal. If with a speed of 90 km/hr the train crosses the platform in one minute, then the length of the train (in metres) is :
(1) 500 (2) 600
(3) 750 (4) 900

2. A train passes two bridges of lengths 800 m and 400 m in 100 seconds and 60 seconds respectively. The length of the train is :
(1) 80 m (2) 90 m
(3) 200 m (4) 150 m

3. A train 300 metres long is running at a speed of 25 metres per second. It will cross a bridge of 200 metres in
(1) 5 seconds (2) 10 seconds
(3) 20 seconds (4) 25 seconds

4. A train 800 metres long is running at the speed of 78 km/hr. If it crosses a tunnel in 1 minute, then the length of the tunnel (in metres) is :
(1) 77200 (2) 500
(3) 1300 (4) 13

5. A train is moving at a speed of 132 km/hour. If the length of the train is 110 metres, how long will it take to cross a railway platform 165 metres long?
(1) 5 seconds (2) 7.5 seconds
(3) 10 seconds (4) 15 seconds

6. A train takes 18 seconds to pass through a platform 162 m long and 15 seconds to pass through another platform 120 m long. The length of the train (in m) is :
(1) 70 (2) 80
(3) 90 (4) 105

7. A train, 150 m long, takes 30 seconds to cross a bridge 500 m long. How much time will the train take to cross a platform 370 m long ?
(1) 36 secs (2) 30 secs
(3) 24 secs (4) 18 secs

8. A 120 metre long train is running at a speed of 90 km per hour. It will cross a railway platform 230 m long in
(1) 4.8 seconds (2) 9.2 seconds
(3) 7 seconds (4) 14 seconds

9. A train travelling at a speed of 30 m/sec crosses a platform, 600 metres long, in 30 seconds. The length (in metres) of train is
(1) 120 (2) 150
(3) 200 (4) 300

10. A train with a uniform speed passes a platform, 122 metres long, in 17 seconds

and a bridge, 210 metres long, in 25 seconds. The speed of the train is
(1) 46.5 km/hour (2) 37.5 km/hour
(3) 37.6 km/hour (4) 39.6 km/hour

11. A train running at 7/11 of its own speed reached a place in 22 hours. How much time could be saved if the train would run at its own speed?
(1) 14 hours (2) 7 hours
(3) 8 hours (4) 16 hours

12. A man with 3/5 of his usual speed reaches the destination 2 and half hours late. Find his usual time to reach the destination.
(1) 4 hours (2) 3 hours
(3) 3.75 hours (4) 4.5 hours

13. A car travelling with 5/7 of its usual speed covers 42 km in 1 hour 40 min 48 sec. What is the usual speed of the car?
(1) 17.8 km/hr (2) 35 km/hr
(3) 25 km/hr (4) 30 km/hr

14. Walking at three-fourth of his usual speed, a man covers a certain distance in 2 hours more than the time he takes to cover the distance at his usual speed. The time taken by him to cover the distance with his usual speed is
(1) 4.5 hours (2) 5.5 hours
(3) 6 hours (4) 5 hours

15. By walking at 3/4 of his usual speed, a man reaches his office 20 minutes later than his usual time. The usual time taken by him to reach his office is
(1) 75 minutes (2) 60 minutes
(3) 40 minutes (4) 30 minutes

16. If a man walks 20 km at 5 km/ hr, he will be late by 40 minutes. If he walks at 8 km/hr, how early from the fixed time will he reach?
(1) 15 minutes (2) 25 minutes
(3) 50 minutes (4) 90 minutes

17. If a man reduces his speed to 2/ 3, he takes 1 hour more in walking a certain distance. The time (in hours) to cover the distance with his normal speed is :
(1) 2 (2) 1
(3) 3 (4) 1.5

18. A student rides on bicycle at 8 km/hour and reaches his school 2.5 minutes late. The next day he increases his speed to 10 km/ hour and reaches school 5 minutes early. How far is the school from his house ?
(1) 5/8km (2) 8 km
(3) 5 km (4) 10 km

19. A man covered a certain distance at some speed. Had he moved 3 km per hour faster, he would have taken 40 minutes less. If he had moved 2 km per hour slower, he would have taken 40 minutes more. The distance (in km) is :
(1) 20 (2) 35
(3) 36.6 (4) 40

20. If a train runs at 40 km/hour, it reaches its destination late by 11 minutes. But if it runs at 50 km/ hour, it is late by 5 minutes only. The correct time (in minutes) for the train to complete the journey is
(1) 13 (2) 15
(3) 19 (4) 21

21. A student walks from his house at a speed of 2.5 km per hour and reaches his school 6 minutes late. The next day he increases his speed by 1 km per hour and reaches 6 minutes before school time. How far is the school from his house?
(1)5/4km (2) 7/4km
(3) 9/4 km (4) 11/4 km

22. A boy is late by 9 minutes if he walks to school at a speed of 4 km/hour. If he walks at the rate of 5 km/hour, he arrives 9 minutes early. The distance to his school is
(1) 9 km (2) 5 km
(3) 4 km (4) 6 km

23. A car can cover a certain distance in 4.5 hours. If the speed is increased by 5 km/hour, it would take 1/2 hour less to cover the same distance. Find the slower speed of the car.
(1) 50 km/hour (2) 40 km/hour
(3) 45 km/hour (4) 60 km/hour

24. Shri X goes to his office by scooter at a speed of 30km/h and reaches 6 minutes earlier. If he goes at a speed of 24 km/h, he reaches 5 minutes late. The distance of his office is
(1) 20 km (2) 21 km
(3) 22 km (4) 24 km

25. Walking at 5 km/hr a student reaches his school from his house 15 minutes early and

walking at 3 km/hr he is late by 9 minutes. What is the distance between his school and his house ?
(1) 5 km (2) 8 km
(3) 3 km (4) 2 km

26. A train 180 m long moving at the speed of 20 m/sec. over-takes a man moving at a speed of 10m/ sec in the same direction. The train passes the man in :
(1) 6 sec (2) 9 sec
(3) 18 sec (4) 27 sec

27. A train 100m long is running at the speed of 30 km/hr. The time (in second) in which it will pass a man standing near the railway line is :
(1) 10 (2) 11
(3) 12 (4) 15

28. How many seconds will a 500 metre long train take to cross a man walking with a speed of 3 km/hr. in the direction of the moving train if the speed of the train is 63 km/hr ?
(1) 25 sec (2) 30 sec
(3) 40 sec (4) 45 sec

29. A train is 125 m long. If the train takes 30 seconds to cross a tree by the railway line, then the speed of the train is :
(1) 14 km/hr (2) 15 km/hr
(3) 16 km/hr (4) 12 km/hr

30. A 120 m long train takes 10 seconds to cross a man standing on a platform. What is the speed of the train?
(1)12 m/sec. (2) 10 m/sec.
(3) 15 m/sec. (4) 20 m/sec.

ANSWER KEY																			
1.	(3)	2.	(3)	3.	(3)	4.	(2)	5.	(2)	6.	(3)	7.	(3)	8.	(4)	9.	(4)	10.	(4)
11.	(3)	12.	(3)	13.	(2)	14.	(3)	15.	(2)	16.	(3)	17.	(1)	18.	(3)	19.	(4)	20.	(3)
21.	(2)	22.	(4)	23.	(2)	24.	(3)	25.	(3)	26.	(3)	27.	(3)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET 3

1. On increasing the speed of a train at the rate of 10 km per hr, 30 minutes is saved in a journey of 100 kms. Find the initial speed of train.
(1)40 kmph (2) 45 kmph
(3)42 kmph (4) 44 kmph

2. A thief is noticed by a policeman from a distance of 200m. The thief starts running and the policeman chases him. The thief and the policeman run at the rate of 10 km./hr and 11 km./hr respectively. What is the distance between them after 6 minutes?
(1) 100 m (2) 190 m
(3) 200 m (4) 150 m

3. A moving train, 66 metres long, overtakes another train of 88 metres long, moving in the same direction in 0.168 minutes. If the second train is moving at 30 km/hr, at what speed is the first train moving?
(1) 85 km/hr. (2) 50 km/hr.
(3) 55 km/hr. (4) 25 km/hr.

4. A constable is 114 metres behind a thief. The constable runs 21 metres and the thief runs 15 metres in a minute. In what time will the constable catch the thief ?
(1) 19 minutes (2) 18 minutes
(3) 17 minutes (4) 16 minutes

5. How much time does a train, 50 m long, moving at 68 km/ hour take to pass another train, 75 m long, moving at 50 km/ hour in the same direction.
(1) 5 seconds (2) 10 seconds
(3) 20 seconds (4) 25 seconds

6. A constable follows a thief who is 200 m ahead of the constable. If the constable and the thief run at speed of 8 km/hour and 7

km/hour respectively, the constable would catch the thief in
(1) 10 minutes (2) 12 minutes
(3) 15 minutes (4) 20 minutes

7. Two trains are running with speed 30 km/hr and 58 km/hr in the same direction. A man in the slower train passes the faster train in 18 seconds. The length (in metres) of the faster train is :
(1) 70 (2) 100
(3) 128 (4) 140

8. Two trains travel in the same direction at the speed of 56 km/h and 29 km/h respectively. The faster train passes a man in the slower train in 10 seconds. The length of the faster train (in metres) is
(1) 100 (2) 80
(3) 75 (4) 120

9. A bus moving at a speed of 45 km/hr overtakes a truck 150 metres ahead going in the same direction in 30 seconds. The speed of the truck is
(1) 27 km/hr (2) 24 km/hr
(3) 25 km/hr (4) 28 km/hr

10. A man covered a certain distance at some speed. Had he moved 3 km per hour faster, he would have taken 40 minutes less. If he had moved 2 km per hour slower, he would have taken 40 minutes more. The distance (in km) is :
(1) 20 (2) 35
(3) 36.6 (4) 40

11. The distance between two cities A and B is 330 km. A train starts from A at 8 a.m. and travels towards B at 60 km/hr. Another train starts from B at 9 a.m. and travels towards A at 75 km/hr. At what time do they meet?
(1) 10 a.m. (2) 10 : 30 a.m.
(3) 11 a.m. (4) 11 : 30 a.m.

12. Two men are standing on opposite ends of a bridge 1200 metres long. If they walk towards each other at the rate of 5m/minute and 10m/minute respectively, in how much time will they meet each other ?
(1) 60 minutes (2) 80 minutes
(3) 85 minutes (4) 90 minutes

13. Two trains, one 160 m and the other 140 m long are running in opposite directions on parallel rails, the first at 77 km an hour and the other at 67 km an hour. How long will they take to cross each other?
(1) 7 seconds (2) 7.5seconds
(3) 6 seconds (4) 10 seconds

14. Two trains are running in opposite direction with the same speed. If the length of each train is 120 metres and they cross each other in 12 seconds, the speed of each train (in km/hour) is
(1) 72 (2) 10
(3) 36 (4) 18

15. Two trains 140 m and 160 m long run at the speed of 60 km/ hour and 40 km/hour respectively in opposite directions on parallel tracks. The time (in seconds) which they take to cross each other, is :
(1) 10 sec. (2) 10.8 sec.
(3) 9 sec. (4) 9.6 sec.

16. A train 'B' speeding with 100 kmph crosses another train C, running in the same direction, in 2 minutes. If the length of the train B and C be 150 metre and 250 metre respectively, what is the speed of the train C (in kmph)?
(1) 75 (2) 88
(3) 95 (4) 110

17. A passenger train running at the speed of 80 kms./hr leaves the railway station 6 hours after a goods train leaves and overtakes it in 4 hours. What is the speed of the goods train?
(1) 32 kmph (2) 50 kmph
(3) 45 kmph (4) 64 kmph

18. Two trains start from a certain place on two parallel tracks in the same direction. The speed of the trains are 45 km/hr. and 40 km/ hr respectively. The distance between the two trains after 45 minutes will be
(1) 2.5 km. (2) 2.75 km.
(3) 3.7 km. (4) 3.75 km.

19. A thief is stopped by a policeman from a distance of 400 metres. When the policeman starts the chase, the thief also starts running. Assuming the speed of the thief as 5 km/h and that of policeman as 9 km/h, how far the thief would have run, before he is over taken by the policeman?
(1) 400 metre (2) 600 metre
(3) 500 metre (4) 300 metre

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20. Two trains of equal length are running on parallel lines in the same direction at 46 km/hour and 36 km/hour. The faster train passes the slower train in 36 seconds. The length of each train is

(1) 72 m (2) 80 m
(3) 82 m (4) 50 m

21. Two trains leave a railway station at the same time. The first rain travels due west and the second train due north. The first train travels 5 km per hr. faster than the second train. If after two hours they are 50 km apart, find the average speed of faster train.

(1) 18 kmph (2) 15 kmph
(3) 20 kmph (4) 25 kmph

22. A carriage driving in a fog passed a man who was walking at the rate of 6 km per hr. in the same direction. He could see the carriage for 4 minutes and it was visible to him up to a distance of 200 metres. Find the speed of the carriage.

(1) 8.75 kmph (2) 8.5 kmph
(3) 8 kmph (4) 9 kmph

23. Two bullets were fired at a place at an interval of 12 minutes. A person approaching the firing point in his car hears the two sounds at an interval of 11 minutes 40 seconds. The speed of sound is 330 metres per second. What is the approximate speed of the car?

(1) 34 kmph (2) 32 kmph
(3) 36 kmph (4) 38 kmph

24. *A* and *B* start simultaneously at 5 km per hr. and 4 km per hr. from *P* and *Q*, 180 kms apart, towards *Q* and *P* respectively. They cross each other at *M* and after reaching *Q* and *P* turn back immediately and meet again at *N*. Find the distance *MN*.

(1) 45 km (2) 40 km
(3) 35km (4) 42 km

25. A car driving in the morning fog passes a man walking at 4 km per hr. in the same direction. The man can see the car for 3 minutes and visibility is upto a distance of 130 metres. Find the speed of the car.

(1) 7.5 kmph (2) 6.6 kmph
(3) 6 kmph (4) 7 kmph

26. Ram starts his journey from Bombay to Pune and simultaneously Mohan starts from Pune to Bombay. After crossing each other they finish their remaining journey in $6\frac{1}{4}$ and 4 hours respectively. What is Mohan's speed if Ram's speed is 20 km per hr.

(1) 28 kmph (2) 24 kmph
(3) 25 kmph (4) 30 kmph

27. Two guns were fired from same place at an interval of 5 minutes but a man sitting in the train going away from the place near the second fire 6 minute after the first if the speed of train is 40 km per hour find the speed of sound

(a) 160 kmph (b) 200kmph
(c) 240 kmph (d) 350 kmph

28. The buses are departed after every 25 minutes but man going towards from the bus deport after every 20 minute get the buses find the speed of message if the speed of man is 30 km per hour?

(a) 120 kmph (b) 200kmph
(c) 150 kmph (d) 180kmph

29. Two men A and B start from Delhi e and Agra at same time towards each other after meeting on the way they cover the remaining journey in in $7\frac{1}{9}$ and $6\frac{1}{4}$ hours respectively find the slower speed if faster speed is 40 kilometre more than the slower

(a) 500 kmph (b) 600 kmph
(c) 650 kmph (d) 720 kmph

30. Two men A and B start from X and Y at same time towards each other after meeting on the way they cover the remaining journey in 9 and 4 hours respectively find the faster speed if slower speed is 40 kilometre

(a) 50 kmph (b) 60 kmph
(c) 65 kmph (d) 72 kmph

ANSWER KEY

1.	(1)	2.	(1)	3.	(1)	4.	(1)	5.	(4)	6.	(2)	7.	(4)	8.	(3)	9.	(1)	10.	(4)
11.	(3)	12.	(2)	13.	(2)	14.	(3)	15.	(2)	16.	(2)	17.	(1)	18.	(4)	19.	(3)	20.	(4)
21.	(3)	22.	(4)	23.	(1)	24.	(2)	25.	(2)	26.	(3)	27.	(3)	28.	(1)	29.	(2)	30.	(2)

PRACTICE SHEET 4

1. In a one-kilometre race A, B and C are the three participants. A can give B a start of 50 m. and C a start of 69 m. The start, which B can allow C is
(1) 17 m. (2) 20 m.
(3) 19 m. (4) 18 m.

2. A runs twice as fast as B and B runs thrice as fast as C. The distance covered by C in 72 minutes, will be covered by A in :
(1) 18 minutes (2) 24 minutes
(3) 16 minutes (4) 12 minutes

3. A is twice as fast runner as B, and B is thrice as fast runner as C. If C travelled a distance in 1 hour 54 minutes, the time taken by B to cover the same distance is
(1) 19 minutes (2) 38 minutes
(3) 51 minutes (4) 57 minutes

4. In a 100m race, Kamal defeats Bimal by 5 seconds. If the speed of Kamal is 18 Kmph, then the speed of Bimal is
(1) 15.4 kmph (2) 14.5 kmph
(3) 14.4 kmph (4) 14 kmph

5. In a race of 1000 m, A can beat B by 100m. In a race of 400 m, B beats C by 40m. In a race of 500m. A will beat C by
(1) 95 m (2) 50 m
(3) 45 m (4) 60 m

6. In a race of 800 metres, A can beat B by 40 metres. In a race of 500 metres, B can beat C by 5 metres. In a race of 200 metres, A will beat C by
(1) 11.9 metre (2) 1.19 metre
(3) 12.7 metre (4) 1.27 metre

7. In a race of 200 metres, B can give a start of 10 metres to A, and C can give a start of 20 metres to B. The start that C can give to A, in the same race, is
(1) 30 metres (2) 25 metres
(3) 29 metres (4) 27 metres

8. A can give 40 metres start to B and 70 metres to C in a race of one kilometre How many metres start can B give to C in a race of one kilometre ?
(1) 30 metre (2) 31.25metre
(3) 31.75 metre (4) 32 metre

9. A jeep is chasing a car which is 5km ahead. Their respective speed are 90 km/hr and 75 km/ hr. After how many minutes will the jeep catch the car?
(1) 18 min. (2) 20 min.
(3) 24 min. (4) 25 min.

10. A is twice as fast as B, and B is thrice as fast as C is. The journey covered by C in 1 and half hours will be covered by A in
(1) 15 minutes (2) 30 minutes
(3) 1 hour (4) 10 minutes

11. It takes 8 hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the car is
(1) 2 : 3 (2) 3 : 2
(3) 3 : 4 (4) 4 : 3

12. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the car is :
(1) 3 : 5 (2) 3 : 4
(3) 4 : 3 (4) 4 : 5

13. A truck covers a distance of 550 metre in one minute where as a bus covers a distance of 33 km in 3/4 hour. Then the ratio of their speeds is
(1) 1 : 3 (2) 2 : 3
(3) 3 : 4 (4) 1 : 4

14. A car travels 80 km. in 2 hours and a train travels 180 km. in 3 hours. The ratio of the speed of the car to that of the train is :
(1) 2 : 3 (2) 3 : 2
(3) 3 : 4 (4) 4 : 3

15. The speeds of three cars are in the ratio of 1 : 3 : 5. The ratio among the time taken by these cars to travel the same distance is
(1) 3 : 5 : 15 (2) 15 : 3 : 5
(3) 15 : 5 : 3 (4) 5 : 3 : 1

16. A man rows a boat 18 kilometres in 4 hours downstream and returns upstream in 12 hours. The speed of the stream (in km per hour) is :

(1) 1 (2) 1.5
(3) 2 (4) 1.75

17. A motorboat in still water travels at a speed of 36 kmph. It goes 56 km upstream in 1 hour 45 minutes. The time taken by it to cover the same distance down the stream will be :
(1) 2 hours 25 minutes
(2) 3 hours
(3) 1 hour 24 minutes
(4) 2 hours 21 minutes

18. A boat running downstream covers a distance of 20km in 2 hrs while it covers the same distance upstream in 5 hrs. Then speed of the boat in still water is
(1) 7 km/hr (2) 8 km/hr
(3) 9 km/hr (4) 10 km/hr

19. A boatman rows 1 km in 5 minutes, along the stream and 6 km in 1 hour against the stream. The speed of the stream is
(1) 3 kmph (2) 6 kmph
(3) 10 kmph (4) 12 kmph

20. A boat covers 24 km upstream and 36 km downstream in 6 hours, while it covers 36 km upstream and 24 km downstream in 6 and half hours. The speed of the current is
(1) 1 km/hr (2) 2 km/hr
(3) 1.5 km/hr (4) 2.5 km/hr

21. A boat goes 6 km an hour in still water, but takes thrice as much time in going the same distance against the current. The speed of the current (in km/hour) is :
(1) 4 (2) 5
(3) 3 (4) 2

22. In a fixed time, a boy swims double the distance along the current that he swims against the current. If the speed of the current is 3 km/hr, the speed of the boy in still water is
(1) 6 km/hr (2) 9 km/hr
(3) 10 km/hr (4) 12 km/hr

23. A man can row 30 km downstream and return in a total of 8 hours. If the speed of the boat in still water is four times the speed of the current, then the speed of he current is
(1) 1 km/hour (2) 2 km/hour
(3) 4 km/hour (4) 3 km/hour

24. A person can row 7.5 km an hour in still water and he finds that it takes him twice as long to row up as to row down the river. The speed of the stream is :
(1) 2 km/hr (2) 3 km/hr
(3) 2.5 km/hr (4) 3.5 km/hr

25. A man can row at a speed of 4.5 km/hr in still water. If he takes 2 times as long to row a distance upstream as to row the same distance downstream, then, the speed of stream (in km/hr) is
(1) 1 (2) 1.5
(3) 2 (4) 2.5

26. A man can row at 5 kmph. in still water. If the velocity of current is 1 kmph. and it takes him 1 hour to row to a place and come back, how far is the place ?
(1) 2.5 km (2) 3 km
(3) 2.4 km (4) 3.6 km

27. The speed of a motor-boat is that of the current of water as 36 : 5. The boat goes along with the current in 5 hours 10 minutes. It will come back in
(1) 5 hours 50 minutes
(2) 6 hours
(3) 6 hours 50 minutes
(4) 12 hours 10 minutes

28. A man goes downstream with a boat to some destination and returns upstream to his original place in 5 hours. If the speed of the boat in still water and the stream are 10 km/hr and 4 km/ hr respectively, the distance of the destination from the starting place is
(1) 16 km (2) 18 km
(3) 21 km (4) 25 km

29. Two boats A and B start towards each other from two places, 108 km apart. Speed of the boat A and B in still water are 12km/hr and 15km/hr respectively. If A proceeds down and B up the stream, they will meet after.
(1) 4.5 hours (2) 4 hours
(3) 5.4 hours (4) 6 hours

30. A man can row 6 km/h in still water. If the speed of the current is 2 km/h, it takes 3 hours more in upstream than in the downstream for the same distance. The distance is
(1) 30 km (2) 24 km
(3) 20 km (4) 32 km

ANSWER KEY																			
1.	(2)	2.	(4)	3.	(2)	4.	(3)	5.	(1)	6.	(1)	7.	(3)	8.	(2)	9.	(2)	10.	(1)
11.	(3)	12.	(2)	13.	(3)	14.	(1)	15.	(1)	16.	(2)	17.	(3)	18.	(1)	19.	(1)	20.	(2)
21.	(1)	22.	(2)	23.	(2)	24.	(3)	25.	(2)	26.	(3)	27.	(3)	28.	(3)	29.	(2)	30.	(2)

CDS PYQ

1. A person travels a certain distance at 3 km/h and reaches 15 min late. If he travels at 4 km/h, he reaches 15 min earlier. The distance he has to travel is

[CDS 2013(I)]

(a) 4.5 km (b) 6 km
(c) 7.2 km (d) 12 km

2. If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then the average speed (in km/h) is

[CDS 2013(I)]

(a) $\frac{(x+y)}{2}$ (d) $\sqrt{xy}$
(c) $\frac{2xy}{x+y}$ (d) $\frac{(x+y)}{xy}$

3. A sailor sails a distance of 48 km along the flow of a river in 8 h. If it takes 12 h to return the same distance, then the speed of the flow of the river is

[CDS 2013(I)]

(a) 0.5 km/h (b) 1 km/h
(c) 1.5 km/h (d) 2 km/h

4. A train running at the speed of 72 km/h goes past a pole in 15 s. What is the length of the train.

[CDS 2013(II)]

(a) 150 m (b) 200 m
(c) 300 m (d) 350 m

5. Two cars A and B start simultaneously from a certain place at the speed of 30 km/h and 45 km/h, respectively. The car B reaches the destination 2 h earlier than A. What is the distance between the starting point and destination?

[CDS 2013(II)]

(a) 90 km (b) 180 km
(c) 270 km (d) 360 km

6. A man cycles with a speed of 10 km/h and reaches his office at 1: 00 pm. However, when he cycles with a speed of 15 km/h, he reaches his Office at 11 00 am. At what speed should he cycle, so that he reaches his office at 12 noon?

[CDS 2013(II)]

(a) 12.5 km/h (b) 12 km/h
(c) 13 km/h (d) 13.5 km/h

7. A train takes 9 s to cross a pole. If the speed of the train is 48 km/h, the length of the train is

[CDS 2014(I)]

(a) 150 m (b) 120m
(c) 90 m (d) 80 m

8. A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of the train?

[CDS 2014(II)]

(a) 50 m (b) 100m
(c) 150 m (d) 200 m

9. A train travels at a speed of 40 km/h and another train at a speed of 20 m/s. What is the ratio of speed of the first train to that of the second train?

[CDS 2014(II)]

(a) 2:1 (b) 5:9
(c) 5:3 (d) 9:5

10. The distance between two points (A and B) is 110 km. X starts running from point A at a 8peed of 60 km/h and Y starts running from point B at a speed of 40 km/h at the same time. They meet at a point C, somewhere on the line OB. What is the ratio of AC to BC?

[CDS 2014(II)]

(a) 3: 2 (b) 2:3
(c) 3:4 (d) 4:3

11. A man rides one-third of the distance from A to Bat the rate of x km/h and the remainder at the rate of 2y km/h. If he had travelled at a uniform rate of 6z km/h, then he could have ridden from A to B and back again in the same time. Which one of the following is correct?

[CDS 2014(II)]

(a) z = x + y (b) 3Z= x+y
(c) $\frac{1}{z}=\frac{1}{x}+\frac{1}{y}$ (d) $\frac{1}{2z}=\frac{1}{x}+\frac{1}{y}$

12. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/h and the time of flight increased by 30 min. The duration of the flight is

[CDS 2015(I)]

(a) 1h (b) 2h
(c) 3h (d) 4h

13. With a uniform speed, a car covers a distance in 8 h. Had the speed been increased by 4 km/h, the same distance could have been covered in 7 h and 30 min. What is the distance covered?

[CDS 2015(I)]

(a) 420 km (b) 480 km
(c) 520 km (d) 640 km

14. A car travels the first one-third of a certain distance with a speed of 10 km/h, the next one-third distance with a speed of 20 km/h and the last one-third distance with a speed of 60 km/h. The average speed of the car for the whole journey is

[CDS 2015(I)]

(a) 18 km/h (b) 24 km/h
(c) 30 km/h (d) 36 km/h

15. A man rows 32 km downstream and 14 km upstream, and he takes 6h to cover each distance. What is the speed of the current?

[CDS 2015(I)]

(a) 0.5 km/h (b) 1 km/h
(c) 1.5 km/h (d) 2 km/h

16. A runs $1\frac{2}{3}$ times as fast as B. If A gives B a start of 80 m, how far must the winning post from the starting point be so that A and B might reach it at the same time?

[CDS 2015(I)]

(a) 200 m (b) 300 m
(c) 270 mn (d) 160 m

17. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the thief and the policeman chases him. Policeman run at the speed of 10 km/h and 11 km/h, respectively. What is the distance between them after 6 min?

[CDS 2015(I)]

(a) 100 m (b) 120m
(c) 150 m (d) 160 m

18. Two persons A and B start simultaneously from two places c km apart and walk in the same direction. If A travels at the rate of p km/h and B travels at the rate of q km/h . then A has travelled before he overtakes B a distance of

[CDS 2015(I)]

(a) $\frac{qc}{p+q}$ (b) $\frac{pc}{p-q}$
(c) $\frac{qc}{p-q}$ (d) $\frac{pc}{p+q}$

19. By increasing the speed ot his car by 15 km/h. a person covers 300 km distance by taking an hour less than before. The original speed of the car was

[CDS 2015(II)]

(a) 45 km/h (b) 50 km/h
(c) 60 km/h (d) 75 km/h

20. Two trains, one is of 121 m in length at the speed of 40 km/h and the other is of 99 m in length at the speed of 32 km/h are running in opposite directions. In how much time will they be completely clear from each other from the moment they meet?

[CDS 2015(II)]

(a) 10 s (b) 11 s
(c) 16 s (d) 21 s

21. The speeds of three buses are in the ratio 2:3:4. The time taken by these buses to travel the same distance will be in the ratio.

[CDS 2015(II)]

(a) 2:3:4 (b) 4:3: 2
(c) 4:3:6 (d) 6:4:3

OFFICIUM SUPREMUM

22. Two trains are moving in the same direction at 1.5 km/min and 60 km/h, respectively. A man in the faster train observes that it takes 27 s to cross the slower train. The length of the slower train is

[CDS 2015(II)]

(a) 225 m (b) 230 m
(c) 240 m (d) 250 m

23. In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race?

[CDS 2015(II)]

(a) 2 (b) 2.04
(c) 2.08 (d) 3.2

24. Three athletes run a 4 km race. Their speeds are in the ratio 16:15:11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third position is

[CDS 2015(II)]

(a) 1000 m (b) 800 m
(c) 750 m (d) 600 m

25. A motorboat, whose speed is 15 km/h in still water goes 30 km downstream and comes back a total of 4h and 30 min. The speed of the stream is

[CDS 2015(II)]

(a) 4 km/h (b) 5 km/h
(c) 6 km/h (d) 10 km/h

26. In a race A, B, C take part. A beats B by 30m, B beats C by 20m and A beats C by 48m. Which of the following is/are correct?
1. The length of race is 300m.
2. The speeds of A, B, C are in the ratio 50: 45: 42.
Select the correct answer using codes given below

[CDS 2015(II)]

(a) Only I (b) Only 1l
(c) Both | and l (d) Neither I nor II

27. A bike consumes 20 mL of petrol per kilometre if it is driven at a speed in the range of 25-50 km/h and consumes 40 mL of petrol per kilometre at any other speed. How much is consumed by the bike in travelling distance of 50 km, if the bike is driven at a speed of 40 km/h for the first 10 km, at a speed of 60 km/h for the next 30 km and at a speed of 30 km/h for the last 10 km?

[CDS 2016(I)]

(a) 1 L (b) 1.2 L
(c) 1.4 L (d) 1.6 L

28. A passenger train takes 1 h less for a journey of 120 km, if its speed is increased by 10 km/h from its usual speed. What is its usual speed?

[CDS 2016(I)]

(a) 50 km/h (b) 40 km/h
(c) 35 km/h (d) 30 km/h

29. A man walking at 5 km/h noticed that a 225 m long train coming in the opposite direction crossed him in 9s. The speed of the train is

[CDS 2016(I)]

(a) 75 km/h (b) 80 km/h
(c) 85 km/h (d) 90 km/h

30. A cyclist moves non-stop from A to B, a distance of 14 km, at a certain average speed. If his average speed reduces by 1 km/h, then he takes 20 min more to cover the same distance. The original average speed of the cyclist is

[CDS 2016(I)]

(a) 5 km/h (b) 6 km/h
(c) 7 km/h (d) None of these

31. In a race of 1000 m, A beats B by 100 m or 10 s. If they start a race of 1000 m simultaneously from the same point and if B gets injured after running 50 m less than half the race length and due to which his speed gets halved, then by how much time will A beat B?

[CDS 2016(I)]

(a) 65 s (b) 60 s
(c) 50 s (d) 45 s

32. Two men, A and B run a race on a course 0.25 km around. If their speeds are in the ratio 5:4, how often does the winner pass the others.

[CDS 2016(II)]

(a) once (b) Twice
(c) Thrice (d) Four times

33. When the speed of a train is increased by 20% , it takes 20 minutes less to cover the same distance. What is the time taken to cover the same distance with the original speed?

[CDS 2016(II)]

(a) 140 min (b) 120 min
(c) 100 min (d) 80 min

34. A person can row downstream 20 km in 2 hours and upstream 4 km in 2 hours. What is the speed of the current?
[CDS 2016(II)]
(a) 2km/hr (b) 2.5 km/hr
(c) 3 km/hr (d) 4km/hr

35. A train is travelling at 48 km/hr completely crosses another train having half its length and travelling in opposite direction at 42 km/hr in 12 s. It also passes a railway platform in 45 s. What is the length of platform?
[CDS 2016(II)]
(a) 600 m (b) 400 m
(c) 300 m (d) 200 m

36. The speeds of three cars are in the ratio 2 : 3 : 4 . What is the ratio between the times taken by these cars to travel the same distance?
[CDS 2016(II)]
(a) 4 : 3 : 2 (b) 2 : 3 : 4
(c) 4 : 3 : 4 (d) 6 : 4 : 3

37. A motorist travels to a place 150km away at an average speed of 50 km/hour and returns at 30km/hr. What is the average speed of the whole journey?
[CDS 2016(II)]
(a) 35 km/hour (b) 37 km/hour
(c) 37.5 km/hour (d) 40 km/hour

38. In a 100m race. A runs at a speed of 5/3 m/s. If A gives a start of 4 m to B and still beats him by 12 seconds, What is the speed of B?
[CDS 2017(I)]
(a) 5/4 m/s (b) 7/5 m/s
(c) 4/3 m/s (d) 6/5 m/s

39. A passenger train departs from delhi at 6 pm from Mumbai. At 9 pm, an express train, whose average speed exceeds that of the passenger train by 15km/hr leaves Mumbai for Delhi. Two trains meet each other mid-route. At what time do they meet, given that the distance between the cities is 1080 km?
[CDS 2017(I)]
(a) 4 pm (b) 2 am
(c) 12 mid night (d) 6 am

40. A 225m long train is running at a speed of 30 km/hr. How much time does it take to cross a man running at 3 km/hr in the same direction?
[CDS 2017(I)]
(a) 40 sec (b) 30 sec
(c) 25 sec (d) 15 sec

41. A thief is spotted by a policeman from a distance of 100m. When the policeman starts the chase, the thief also starts running. If the speed of the thief is 8km/hr and that of the policeman is 10 km/hr, then how far will the thief have to run before the overtaken?
[CDS 2017(I)]
(a) 200m (b) 300m
(c) 400m (d) 500m

42. If a train crosses a km-stone in 12 seconds, how long will it take to cross 91 km-stones completely if its speed is 60 km/hr?
[CDS 2017(II)]
(a) 1hr 30 min
(b) 1 hr 30 min 12 sec
(c) 1 hr 51 min
(d) 1 hr 1 min 3 sec

43. In a 100m race, A runs at 6km/hr. If a gives B a start of 8m and still beats him by 9 seconds. What is the speed of B?
[CDS 2017(II)]
(a) 4.6 km/hr (b) 4.8 km/hr
(c) 5.2 km/hr (d) 5.4 km/hr

44. A boy went to school at a speed of 12km/hr and returned to his house at a speed of 8 km/hr. If he has taken 50 minutes for the whole journey, what was the total distance walked?
[CDS 2017(II)]
(a) 4km (b) 8km
(c) 16km (d) 20 km

45. A man rows down a river at 18 km in 4 hours with the stream and returns in 10 hours.
Consider the following statements
1. The speed of man against the stream is 1.8 km/hr
2. The speed of the man in still water is 3.15 km/hr
3. The speed of the stream is 1.35 km/hr.
Which of the above statements are correct?
[CDS 2017(II)]
(a) 1 and 2 (b) 2 and 3

(c) 1 and 3 (d) 1,2 and 3

46. A man travelled 12km at a speed of 4 km/hr and further 10 km at a speed of 5 km/hr. What was his average speed?

[CDS 2017(II)]

(a) 4.4 km/hr (b) 4.5 km/hr
(c) 5.0 km/hr (d) 2.5 km/hr

47. A train moving with a speed of 60 km per hour crosses an electric pole in 30 second what is the length of the train in metres?

[CDS 2018(I)]

(a) 300 (b) 400
(c) 500 (d) 600

48. A passenger train and a goods train are running in the same direction on parallel railway tracks, if the passenger train now takes three times as long to pass the goods train as when they are running in opposite directions, then what is the ratio of the speed of the passenger train to that of the goods train (assume that the trains run at uniform speed)

[CDS 2018(I)]

(a) 2:1 (b) 3:2
(c) 4:3 (d) 1:1

49. A man can row at a speed of X kilometre per hour in still water if in a stream which is flowing at a speed of Y kilometre per hour it takes him *z* hours to row to a place and back then what is the distance between the two places ?

[CDS 2018(I)]

(a) $\frac{z(x^2-y^2)}{2y}$ (b) $\frac{z(x^2-y^2)}{2x}$

(c) $\frac{(x^2-y^2)}{2zx}$ (d) $\frac{z(x^2-y^2)}{x}$

50. A car has an average speed of 60 km/hr while going from Delhi to Agra and has an average speed of y km/hr while returning to Delhi from Agra by travelling the same distance if the average speed of the car for the whole journey is 40 km per hour then what is the value of Y

[CDS 2018(I)]

(a) 30 km/hr (b) 35km/hr
(c) 40 km/hr (d) 45 km/hr

51. A cyclist covers his first 20 km at an average speed of 40 kmph, another 10 km at an average speed of 10 kmp/h and the last 30 km at an average speed of 40 kmp/h. Then, the average speed of the entire journey is

[CDS 2018(II)]

(a) 20 km/h (b) 26.67 km/h
(c) 28.24 km/h (d) 30 km/h

52. A train 100 m long passes a platform 100 m long in 10 s. The speed of the train is

[CDS 2018(II)]

(a) 36 km/h (b) 45 km/h
(c) 54 km/h (d) 72 km/h

53. In a race of 1000 m, A beats B by 150 m, while in another race of 3000 m, C beats D by 400 m. Speed of B is equal to that of D. (Assume that A, B, C and D run with uniform speed in all the events). If A and C participate in a race of 6000 m, then which one of the following is correct

[CDS 2018(II)]

(a) A beats C by 250 m
(b) C beats A by 250 m
(c) A beats C by 115.38 m
(d) C beats A by 115.38 m

54. A thief steals a car parked in a house and goes away with a speed of 40 km/h. The theft was discovered after half an hour and immediately the owner sets off in another car with a speed of 60km/h. When will the owner meet the thief?

[CDS 2018(II)]

(a)55 km from the owner's house and one hour after the theft
(b)60 km from the owner's house and 1.5 h after the theft
(c)60 km from the owner's house and 1.5 h after the discovery of of the theft
(d)55 km from the owner's house and 1.5 h after the theft

55. Three cars A, B and C started from a point at 5 pm, 6 pm and 7 pm, respectively uniform speeds of 60 km/h, 80 km/h and x km/h, respectively in the same direction. If all the three travelled at and meet at another point at the same instant during their journey, then what is the value of x?

[CDS 2019(I)]

(a) 120 (b) 110
(c) 105 (d) 100

56. Radha and Hema are neighbours and study in the same school. Both of them use bicycles to go to the school. Radha's speed is

8 km/h whereas Hema's speed is 10 km/h. Hema takes 9 min less than Radha to reach the school. How far is the school from the locality of Radha and Hema?

[CDS 2019(I)]

(a) 5 km (b) 5.5 km
(c) 6 km (d) 6.5 km

57. A plane is going in circles around an airport. The plane takes 3 min to complete one round. The angle of elevation of the plane from a point P on the ground at time t is equal to that at time (t +30) s. At time (t+x) s, the plane flies vertically above the point P. What is x equal to?

[CDS 2019(I)]

(a) 75s (b) 90 s
(c) 105 s (d) 135 s

58. A race has three parts. The speed and time required to complete the individual parts for a runner is displayed on the following chart

	Part 1	Part 2	Part 3
Speed(km/hr)	9	8	7.5
Time (Min)	50	80	100

What is the average speed of this runner

[CDS 2019(I)]

(a) 8.17 km/h (b) 8 km/h
(c) 7.80 km/h (d) 7.77 km/h

59. It takes 11 h for a 600 km journey if 120 km is done by train and the rest by car. It takes 40 min more if 200 km are covered by train and the rest by car. What is the ratio of speed of the car to that of the train?

[CDS 2019(II)]

(a) 3:2 (b) 2:3
(c) 3:4 (d) 4:3

60. The ratio of speeds of X and Y is 5:6. If Y allows X a start of 70 m in a 1.2 km race, then who will win the race and by what distance ?

[CDS 2020(I)]

(a) X wins the race by 30 m
(b) Y wins the race by 90 m
(c) Y wins the race by 130 m
(d) The race finishes in a dead heat

61. A train takes two hours less for at journey of 300 km if its speed, is increased by 5 km/hr from its usual speed. What is its usual speed?

[CDS 2020(I)]

(a) 50 km/hr (b) 40 km/hr
(c) 35 km/hr (d) 25 km/hr

62. In covering certain distance, the average speeds of X and Y are in the ratio 4:5. If X takes 45 minutes more than Y to reach the destination, then what is the time taken by Y to reach the destination?

[CDS 2020(I)]

(a) 135 minutes (b) 150 minutes
(c) 180 minutes (d) 225 minutes

63. A train of length 110 m is moving at a uniform speed of 132 km/hr. the time required to cross a bridge of length 165m is

[CDS 2020(II)]

(a) 6.5 sec (b) 7 sec
(c) 7.5 sec (d) 8.5 sec

64. By increasing the speed of his car by 15 km/hr, a person cover a distance of 300 km by taking an hour less than before. What was the original speed of car?

[CDS 2020(II)]

(a) 45 km/hr (b) 50 km/hr
(c) 60 km/hr (d) 75 km/hr

65. X, Y and Z travel from the same place with uniform speeds 4km/hr, 5km/hr and 6 km/hr respectively. Y starts 2 hours after X. How long after Y must Z start in order that they overtake X at the same instant?

[CDS 2020(II)]

(a) 3/2 hours (b) 4/3 hours
(c) 9/8 hours (d) 11/8 hours

66. A car did a journey in t hours, Had the average speed been x kmph greater, the journey would have taken y hours less. How long was the journey?

[CDS 2020(II)]

(a) x(t-y)ty (b) $x(t-y)ty^{-1}$
(c) $x(t-y)ty^{-2}$ (d) x(t+y)ty

67. When a ball is allowed to fall, the time it takes to fall any distance varies as the square root of the distance and it takes 4 seconds to fall 78.40 m. How long would it take to fall 122.50?

[CDS 2020(II)]

(a) 5 sec (b) 5.5 sec
(c) 6 sec (d) 6.5 sec

68. A train 200 m long passes a platform 100 m long in 10 seconds. What is the speed of train ?

[CDS 2021(I)]

(a) 40 m/s (b) 30 m/s
(c) 25 m/s (d) 20 m/s

69. A train travels 600 km in 5 hours and the next 900 km in 10 hours. What is the average speed of the train?

[CDS 2021(I)]

(a) 80 km/hr (b) 90 km/hr
(c) 100 km/hr (d) 120 km/hr

70. Walking at 4/5 of his usual speed, a man is 12 minute late for his office. What is the usual time taken by him to cover that distance?

[CDS 2021(I)]

(a) 48 min (b) 50 min
(c) 54 min (d) 60 min

71. A boat has speed 30km/hr in still water. It goes 60km downstream and comes back in 9/2 hours. What is the speed of the stream?

[CDS 2021(II)]

(a) 5km/hr (b) 8km/hr
(c) 10km/hr (d) 12km/hr

72. A person rode one-third of a journey at 60km/hr, one third at 50km/hr and the rest at 40km/hr. Had the person ridden half of the journey at 60km/hr and the rest at 40km/hr, he would have taken 4 minute longer to complete the journey. What is the person ride?

[CDS 2021(II)]

(a) 180 km (b) 210 km
(c) 240 km (d) 300 km

73. A train X takes 2 hours less than a train Y to cover a distance of 192 km between two cities. Their average speed differ by 16km/her. How long does the faster train, take to cover the journey?

[CDS 2022(I)]

(a) 3 hours (b) 4 hours
(c) 5 hours (d) 6 hours

74. A car travels from A to B at a speed of 40 km /hr. travels back from B to A at a speed of 30km/hr and again goes from A to B at a speed of 60 km/hr. What is the average speed of the car?

[CDS 2022(I)]

(a) $\frac{130}{3}$ km/hr (b) 42 km/hr
(c) 40km/hr (d) $\frac{125}{3}$ km/hr

75. A man walks at an average speed of 3km/her from his residence and reaches office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence and office?

[CDS 2022(I)]

(a) 6km (b) 8km
(c) 10km (d) 12km

76. A car takes p minutes to travel a distance of 350km with an average speed of u km/hr. Another can takes q minutes to travel the same distance with an average speed of v km/hr. If u – v = 5 and q – p = 140, then what is the value of u?

[CDS 2022(I)]

(a) 35 (b) 30
(c) 25 (d) 20

77. In a flight of 2800 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 100 km/hr and time of flight increased by 30 minutes. What was the original average speed of the aircraft?

[CDS 2022 (II)]

(a) 700km/hr (b) 750km/hr
(c) 800km/hr (d) 900km/hr

78. A person X starts from a place A and another person Y start simultaneously from another place B which is d km away from A. They walk in the same direction. X walks at an average speed of u km/hr and Y walks at an average speed of v km/hr. How far will X have walked before he overtakes Y?

[CDS 2022 (II)]

(a) $\frac{ud}{(u-v)}$ (b) $\frac{vd}{(u-v)}$
(c) $\frac{(ud-vd)}{(u-v)}$ (d) $\frac{(ud+vd)}{(u+v)}$

79. X takes 3 hours longer than Y to walk 30 km. If X doubles his speed, he takes 2 hours less than Y. What is the speed of Y?

[CDS 2022 (II)]

(a) 3km/hr (b) 4km/hr
(c) $4\frac{2}{7}$ km/hr (d) $4\frac{3}{7}$ km/hr

80. There are two station X and Y, 1320 km apart. A train starts from station X at 6 a.m. and moves at an average speed of 60 km/hr. At 2 p.m. another train starts from Y towards X and moves at an average speed of 80km/hr. When do they meet?

[CDS 2022 (II)]

(a) 6p.m. (b) 7p.m.
(c) 8p.m (d) 9p.m.

81. The speed of a boat in still water is 15 km/her. If it can travel 42 km downstream and 28 km upstream in the same time, then what is the speed of the stream?

[CDS – 2023 (1)]

(a) 2.5 km/hr (b) 3 km/hr
(c) 4.5 km/hr (d) 6 km/hr

82. The time taken by a train to cross a man travelling in another train is 10 seconds, when the other train is travelling in the opposite direction. However, it takes 20 seconds, if both the trains are travelling in the same direction. The length of the first train is 200 m and that of the second train is 150m. What is the speed of the first train?

[CDS – 2023 (1)]

(a) 60km/hr (b) 56km/hr
(c) 54km/hr (d) 52km/hr

83. A train completely overtakes two persons, walking in the same direction with speeds 3 km/hr and 4 km/hr in 9 seconds and 75/9 seconds respectively. What is the length of the train?

[CDS-2023 (2)]

(a) 60m (b) 62.5m
(c) 55m (d) 67.5m

84. A man walks at an average speed of 3km/hr from his home and reaches office 40 minutes early. If he walks at an average speed of 2km/hr, he would reach office 40 minutes late. What is the distance between his home and office?

[CDS-2023 (2)]

(a) 6km (b) 8km
(c) 10km (d) 12km

Consider the following for the next three (03) items that follow:
Two trains A and B started from stations P and Q respectively towards each other. Train A started at 7 p.m. at a speed of 60 km/hr and train B started at 4 am (next day) at a speed of 90 km/hr. The distance between the two station P and Q is 800 km

85. How far from station Q will the two trains meet?

[CDS – 2024 (1)]

(a) 140 km (b) 144 km
(c) 156 km (d) 504 km

86. At what time will the two trains meet?

[CDS – 2024 (1)]

(a) 5.28 a.m. (b) 5.44 a.m.
(c) 4.56 a.m. (d) 6.25 a.m.

87. If the lengths of the two trains A and B are 400 m and 500 m respectively, then what is the time taken by them to cross each other?

[CDS – 2024 (1)]

(a) 21.6 seconds (b) 18.2 seconds
(c) 17.4 seconds (d) 15.4 seconds

88. Two trains A and B leave Delhi for Hyderabad at 7:00 a.m. and 7:50 a.m. on the same day and travel at 80 kmph and 100 kmph respectively. After how many kilometers from Delhi will the two trains be together?

[CDS – 2024 (1)]

(a) $\frac{200}{3}$ km (b) 100 km
(c) $\frac{400}{3}$ km (d) $\frac{1000}{3}$ km

89. The speeds of four cars are 2u, 3u, 4u and xu and the time taken by them to cover the same distance is xt, 4t, 3t and 2t respectively, where x, u, t are real numbers. What is the value of x?

[CDS – 2024 (1)]

(a) 8 (b) 6
(c) 5 (d) 2

ANSWER KEY																			
1.	(b)	**2.**	(c)	**3.**	(b)	**4.**	(c)	**5.**	(b)	**6.**	(b)	**7.**	(b)	**8.**	(d)	**9.**	(b)	**10.**	(a)
11.	(c)	**12.**	(a)	**13.**	(b)	**14.**	(a)	**15.**	(c)	**16.**	(a)	**17.**	(a)	**18.**	(c)	**19.**	(c)	**20.**	(b)
21.	(d)	**22.**	(a)	**23.**	(b)	**24.**	(a)	**25.**	(b)	**26.**	(c)	**27.**	(d)	**28.**	(d)	**29.**	(c)	**30.**	(c)
31.	(a)	**32.**	(b)	**33.**	(c)	**34.**	(d)	**35.**	(a)	**36.**	(c)	**37.**	(c)	**38.**	(b)	**39.**	(d)	**40.**	(b)
41.	(c)	**42.**	(b)	**43.**	(b)	**44.**	(b)	**45.**	(d)	**46.**	(a)	**47.**	(c)	**48.**	(a)	**49.**	(b)	**50.**	(c)
51.	(b)	**52.**	(d)	**53.**	(c)	**54.**	(b)	**55.**	(a)	**56.**	(c)	**57.**	(c)	**58.**	(b)	**59.**	(a)	**60.**	(c)
61.	(d)	**62.**	(c)	**63.**	(c	**64.**	(c)	**65.**	(b)	**66.**	(b)	**67.**	(b)	**68.**	(b)	**69.**	(c)	**70.**	(a)
71.	(c)	**72.**	(c)	**73.**	(b)	**74.**	(c)	**75.**	(b)	**76.**	(b)	**77.**	(c)	**78.**	(a)	**79.**	(c)	**80.**	(c)
81.	(b)	**82.**	(c)	**83.**	(b)	**84.**	(b)	**85.**	(c)	**86.**	(b)	**87.**	(a)	**88.**	(d)	**89.**	(b)		

CAPF PYQ

1. A vehicle with mileage 15 km per litre contains 2 L of fuel. The vehicle gets some defect as a result of which 5 L of fuel gets wasted per hour when the engine is on. With what minimum speed the vehicle has to move to travel 20 km with the existing amount of fuel, if it travels with a uniform speed?

[CAPF 2016]

(a) 100 km per hour
(b) 120 km per hour
(c) 150 km per hour
(d) 200 km per hour

2. Two men set out at the time to walk towards each other from points A and B, 72 km apart. The first man walks at the speed of 4 kmph while the second walks 2 km in the first hour, 2.5 km in second hour , 3 km in the third hour and so on. The two men will meet

[CAPF 2017]

(a) in 8 hours
(b) nearer to A then B
(c) nearer to B then A
(d) midway between A and B

3. A runner's average speed reduces by 25% every hour. If he runs 16km in the first hour and he runs for 3 hours, then what is his overall average speed?

[CAPF 2022]

(a) 12km/hr (b) 12.33km/hr
(c) 10.33km/hr (d) 13km/hr

4. A car travels $\frac{3}{4}$th of the distance at a speed of 60 km/hr and the remaining $\frac{1}{4}$th of the distance at a speed of v km/hr. If the average speed for the full journey is 50km/hr, then the value of v is:

[CAPF 2022]

(a) 40 (b) 30
(c) 100/3 (d) 35

5. Two friends 10km apart start running towards each other at speeds of 10km/hr and 14km/hr respectively. After how much time will they meet each other?

[CAPF 2022]

(a) 20 minutes (b) 25 minutes
(c) 28 minutes (d) 30 minutes

ANSWER KEY																			
1.	(c)	**2.**	(d)	3.	(b)	4.	(c)	5.	(b)										